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Aryabhatta Knowledge University, Patna

B.Tech 5th Semester Examination, 2021

Time: 03 Hours

Code: 101507 ***Subject: Transportation Engineering*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven): **[14]**

(a) In a national highway, the minimum width of road and shoulder on each side are respectively.
(i) 10 m and 2 m
(ii) 8 m and 2 m
(iii) 10 m and 1.5 m
(iv) 8 m to 1.5 m

(b) The road connecting a city to highway is called
(i) arterial street
(ii) sub-arterial street
(iii) collector street
(iv) local street

(c) The longest road constructed during the time of Shershah Suri was from
(i) Lahore to Kolkata
(ii) Lahore to Delhi
(iii) Lahore to Patna
(iv) Lahore to Agra

(d) The sequence of survey in a road project is
(i) topographic, reconnaissance, preliminary and detailed
(ii) reconnaissance, topographic, preliminary and detailed
(iii) preliminary, reconnaissance, topographic and detailed
(iv) topographic, preliminary, reconnaissance and detailed

(e) In preliminary survey of a road project, in hilly region cross sections are taken generally at
(i) 20 m
(ii) 30 m
(iii) 50 m
(iv) 100 m

(f) Ascending order according to width is
(i) right of way, formation width, carriageway
(ii) formation width, carriageway, right of way
(iii) right of way, carriageway, formation width
(iv) carriageway, formation width, right of way

(g) The order in which a road is built is
(i) sub-soil, base, sub-grade, sub-base
(ii) base, sub-soil, sub-grade, sub-base
(iii) sub-base, base, sub-grade, sub-soil
(iv) sub-soil, sub-grade, sub-base, base

(h) The factor-influencing camber is
(i) topography of the area
(ii) type of wearing coat provided
(iii) amount of rainfall
(iv) nature of sub-soil

(i) For earthen roads in heavy rainfall area, common camber is
(i) 1 in 25
(ii) 1 in 33
(iii) 1 in 50
(iv) 1 in 60

(j) Minimum right-of-way required for a village road in open area is
(i) 9 m
(ii) 10 m
(iii) 12 m
(iv) 15 m

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Q.2 Solve both questions : **[14]**

(a) Compare the construction method of Telford and Macadam; bring out the point of differences.

(b) Explain the necessity and objects of highway planning.

Q.3 Write short notes on the following: **[14]**

(a) Nagpur Road Plan

(b) Fact finding surveys

(c) Star and grid system

(d) Saturation system

Q.4 Solve both questions : **[14]**

(a) What are the objects of reconnaissance surveys? Discuss the scope of aerial survey for the purpose.

(b) Find the total width of a pavement on a horizontal curve for a new national highway to be aligned along a rolling terrain with ruling minimum radius. V = 80 kmph, pavement width W = 7.0m, number of lanes = 2, wheel base = 6 m, skid resistance = 0.15 and max super elevation = 0.07.

Q.5 Solve all questions : **[14]**

(a) Explain the role of pavement surface characteristics in highway geometric design.
State the factor affecting friction between pavements and type of vehicle.

(b) Explain total reaction time of driver and the factor on which it depends. Explain PIEV theory.

Q.6 Solve both questions : **[14]**

(a) The deviation angle at a summit curve is 0.05 and the overtaking sight distance is 300 m. Find the length of summit curve required.

(b) Explain the term 'traffic volume'. What are the objects of carrying out traffic volume study?

Q.7 Solve this question : **[14]**

The CBR value of subgrade soil is 8 percent. Calculate the total thickness of flexible pavement using- (a) design curve developed by California State Highway Department; (b) design chart recommended by IRC; (c) design formula developed by the US Corps Engineers.

Q.8 Solve both questions : **[14]**

(a) Explain how climatic variation affects pavement design and performance.

(b) Discuss the advantages and limitations of CBR method of design.

Q.9 Solve both questions : **[14]**

(a) Specify the material required for construction of WBM roads. What are the uses and limitations of this type of road?

(b) Compare the following methods of bituminous road construction:
(i) Central plant mix and road mix method
(ii) Hot mix and cold mix

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